SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1021	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	15.10.2026	**Duration:**	60 minutes

Code of Conduct

I ANGULAKSHMI S (192524402) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ANGULAKSHMI S

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

1. Project Vision and Objectives

1.1 Project Vision

The vision is to develop a distributed, climate-aware irrigation support network that helps agricultural users make timely, zone-specific watering decisions using validated soil and environmental information. The system is intended to reduce avoidable water use and over-saturation, support the preservation of soil health, and continue safe local operation during loss of persistent cloud connectivity.

1.2 Problem to Be Addressed

Conventional timer-based and shallow-sensor approaches can apply water without accounting for deeper soil moisture, nutrient conditions, local weather changes, or differences between field zones. Overwatering can contribute to waterlogging and nutrient movement below the root zone, while underwatering can expose crops to stress. Large fields also create coordination challenges when sensor readings arrive at different times or connections fail. The proposed software must organise these competing needs into understandable, testable work increments.

1.3 Product Objectives

- Collect and associate high-frequency sensor telemetry with the correct field zone and soil depth.
- Present a multi-depth view of moisture and relevant soil and environmental indicators.
- Generate transparent, zone-specific irrigation recommendations from valid readings and configured rules.
- Account for weather context and flag situations in which a recommendation should be held for review.
- Plan for local operation and safe degradation when connectivity or sensor quality is unreliable.
- Maintain a historical record of readings, recommendations, water utilisation, and relevant soil-health indicators.
- Use iterative stakeholder feedback to refine the product while avoiding unsupported claims about yield improvement or microbiome preservation.

1.4 Product Value Statement

For farmers and field supervisors managing multiple irrigation zones, the proposed network provides clearer evidence for irrigation decisions than fixed schedules alone. It links each recommendation to current data, zone context, and configured rules so that the user can understand

why a zone is advised to irrigate, hold, or be reviewed. This supports informed human supervision rather than treating automation as infallible.

1.5 Success Indicators for the Early Prototype

Indicator	How it will be checked in Sprint 1	Evidence
Zone identity	Every sample reading is linked to a registered zone	Demonstration dataset and test results
Data quality visibility	Stale or invalid readings are labelled and excluded from recommendations	Validation scenarios
End-to-end decision transparency	Multi-depth readings are displayed; each recommendation shows status, timestamp, and reason; no physical valve is activated	Acceptance-test record and sprint review

2. Stakeholder Identification and Responsibilities

Stakeholders are identified according to their relationship to irrigation decisions, system operation, agronomic correctness, maintenance, and project governance. Their feedback should be collected throughout the Scrum cycle, not only at final delivery.

Stakeholder	Role / responsibility	Primary concern	Engagement in Sprint 1
Farm owner / farm manager	Defines operational priorities, resource constraints, and reporting needs	Water use, cost, reliability, visibility across zones	Confirms problem priority and reviews the demo
Field supervisor / irrigation operator	Monitors field conditions and acts on recommendations	Clear zone status, safe operation, understandable alerts	Validates zone workflow and exception messages
Agronomist / soil specialist	Advises on crop, soil, moisture interpretation, and soil-health indicators	Agronomic validity and avoidance of misleading thresholds	Reviews rule assumptions and identifies what requires field validation
Product Owner	Maintains product vision, backlog order, and stakeholder alignment	Value, scope control, and acceptance	Clarifies stories, orders backlog, accepts or rejects completed work
Scrum Master / facilitator	Facilitates Scrum events and removes process impediments	Team flow, transparency, collaboration, continuous improvement	Facilitates planning, daily coordination, review, and retrospective
Development team	Designs and produces the increment and tests the stories	Clear requirements, manageable scope, quality	Breaks stories into tasks and delivers the increment
Quality / test responsibility	Defines scenarios and checks acceptance criteria	Repeatable tests, valid/invalid data handling, regression risk	Prepares tests and records evidence; can be a team responsibility

Stakeholder	Role / responsibility	Primary concern	Engagement in Sprint 1
Maintenance / support personnel	Handles user issues, configuration, and operational continuity	Recoverability, documentation, understandable failures	Identifies support information needed for future increments
Environmental / sustainability adviser	Reviews water-conservation and soil-preservation claims	Evidence-based reporting and responsible claims	Checks that outcome measures are defined without overstating results
Data owner / authorised administrator	Defines access rights and appropriate use of farm data	Privacy, integrity, appropriate access	Reviews proposed access and audit needs for later backlog items

2.1 Stakeholder Engagement Principles

The Product Owner represents priorities and confirms acceptance but does not replace agronomic expertise. The development team estimates work collaboratively rather than receiving estimates solely from a manager. Field users should review realistic exception scenarios, and environmental benefits remain hypotheses until measured against an agreed baseline.

2.2 Stakeholder Challenges and Discovery Strategy

Challenge	Potential consequence	Discovery question / response
Differences between zones	Uniform schedules may overwater one zone while another remains dry	Which field differences must be visible before an irrigation decision is made?
Incomplete or stale readings	An apparently precise recommendation may rely on old data	How should the system label uncertain readings and prevent them from being treated as current?
Weather changes	A planned watering action may become unnecessary or require review	Which weather signals should trigger a hold, warning, or updated recommendation?
Limited connectivity	A cloud-dependent process may become unavailable during a network drop	Which actions must continue locally, and which can wait for synchronisation?
Different levels of digital experience	Users may misinterpret technical status messages	Can a user understand the zone state and the reason for a recommendation without specialist training?
Uncertain soil microbiome measurement	A proxy may be mistaken for a direct microbial measurement	Which indicators are measured, which are inferred, and what evidence supports each interpretation?
Operational and cost constraints	An ambitious feature set may exceed time, budget, or training capacity	Which minimum increment can provide useful evidence before investing in physical deployment?

2.3 Applying Design Thinking Within Unit 1

The first sprint should begin with user-centred discovery rather than assumptions about what the dashboard ought to display. Design thinking is applied as a short, practical cycle:

1. Empathise: gather examples from a farm manager and field operator about how watering decisions are currently made and which mistakes are costly.
2. Define: express the need as a problem statement, for example, “The operator needs to distinguish valid, current moisture readings from stale readings before acting on a zone recommendation.”
3. Ideate: consider a small set of alternatives, such as a zone table, depth profile, and a clear recommendation status with a reason.
4. Prototype: prepare a low-cost, reviewable screen or workflow using sample zones and readings.
5. Test: ask representative users to explain what action they would take in valid, stale, and conflicting-data scenarios; record confusion and revise the backlog.

Practical evidence

The workshop should retain short interview notes, prioritisation rationale, acceptance-test results, and review feedback. These artefacts demonstrate that sprint choices reflect stakeholder needs rather than personal preference.

2.4 Scope Boundary for the First Sprint

Sprint 1 is limited to a small software demonstration: zone setup, simulated readings, validation, multi-depth display, and rule-based recommendations. Physical valve control, production-scale distribution, live microbial measurement, predictive yield modelling, and field-verified savings remain future work.

3. Epics and User Stories

Epics group related needs into manageable product areas. User stories describe the value from the perspective of a stakeholder and follow the required format: “As a [user], I want [feature], so that [benefit].” The stories below are written to be independently discussable and estimable; the backlog remains open to refinement as stakeholders provide evidence.

Epic ID / Epic	Story ID	User story	Initial priority
E1 - Field and Zone Management	US-01	As a farm manager, I want to register field zones and their crop labels, so that observations and decisions are associated with the correct area.	Must

Epic ID / Epic	Story ID	User story	Initial priority
E2 - Telemetry Intake	US-02	As a field operator, I want sensor readings to be associated with a zone, depth, and timestamp, so that the soil profile can be reviewed in context.	Must
E2 - Telemetry Intake	US-03	As a field operator, I want stale or invalid readings to be flagged, so that they are not mistaken for reliable current conditions.	Must
E3 - Soil Profile Visibility	US-04	As a farm manager, I want to view moisture readings at different soil depths for each zone, so that surface dryness is not the only basis for a decision.	Must
E4 - Irrigation Decision Support	US-05	As an irrigation operator, I want a rule-based recommendation for each zone with an explanation, so that the next action is understandable and reviewable.	Must
E4 - Irrigation Decision Support	US-06	As an irrigation operator, I want a weather-related hold or warning to be visible, so that imminent weather changes can be considered before irrigation.	Should
E5 - Local Autonomy	US-07	As a field supervisor, I want configured local rules to remain available during a connectivity interruption, so that essential local decisions do not depend on continuous cloud access.	Should
E6 - Audit Trail	US-08	As a farm manager, I want readings, recommendations, and resulting actions to be recorded, so that water use and decision history can be reviewed.	Should
E6 - Audit Trail	US-09	As a sustainability adviser, I want water-use and nutrient-related trends to be summarised by zone, so that potential conservation outcomes can be evaluated against a baseline.	Could
E7 - Crop Stress Assessment	US-10	As an agronomist, I want a zone-level crop-stress indicator based on available observations and weather context, so that areas needing inspection can be prioritised.	Could
E8 - Safe Oversight	US-11	As an authorised operator, I want to hold or override an irrigation recommendation, so that field judgement can be applied when conditions require it.	Should
E8 - Safe Oversight	US-12	As a farm administrator, I want different user permissions for configuration and operational review, so that sensitive settings and records are appropriately controlled.	Should

3.1 Story Quality and Definition of Ready

The story catalogue translates the allocated problem into stakeholder outcomes without prescribing a specific technology stack. Stories are kept small enough to discuss, test, estimate, and prioritise. The first sprint selects five Must-priority stories in dependency order; the remaining stories stay in the Product Backlog for later refinement.

Definition of Ready (used before sprint commitment)

- The user and intended benefit are clear, and the story is linked to a project objective.
- Acceptance conditions can be explained and tested with sample data or a user walkthrough.
- Important dependencies and assumptions have been recorded.

- The Product Owner can explain why the story is prioritised now.
- The development team understands the work sufficiently to estimate it using relative story points.
- The story is sized appropriately for a sprint, or is split into smaller deliverable outcomes.

3.2 Planned Acceptance Summary for Stories Outside Sprint 1

Story	Minimum future acceptance outcome	Why it is not in Sprint 1
US-06 Weather hold	Recommendation indicates the weather condition considered and explains any hold or warning.	Sprint 1 validates the core data-to-recommendation path first; weather integration and policies require stakeholder clarification.
US-07 Local autonomy	A controlled scenario shows configured local rules remain available during a simulated network interruption and records a recovery state.	Requires separate offline/recovery scenarios and verification beyond the first software slice.
US-08 Audit trail	Each recommendation has a timestamp, zone, decision status, input-data reference, and explanation in a reviewable history.	High value, but added after the basic recommendation flow is verified.
US-09 Water/nutrient trends	Reports provide totals and trends for available recorded values and clearly identify missing data.	Requires a sufficiently consistent history and agreed definitions of water and nutrient metrics.
US-10 Crop-stress indicator	Indicator output identifies input assumptions and avoids presenting a prediction as a confirmed diagnosis.	Requires suitable historic and agronomic data before model validity can be established.
US-11 Manual hold/override	An authorised operator can hold a recommendation and the action is visibly acknowledged.	Needs safety rules and authority boundaries to be agreed with farm stakeholders.
US-12 User permissions	Only authorised roles can change field configuration or restricted settings; unauthorised attempts are refused.	Access policy and roles must be confirmed before detailed implementation.

3.3 Dependency Logic

US-01 establishes the zone identity needed by the remaining work. US-02 depends on zones so readings can be assigned correctly. US-03 depends on incoming readings and their timestamps. US-04 depends on valid readings with depth labels. US-05 depends on the zone profile and validation result. This sequence reduces rework and creates a testable end-to-end slice by the end of Sprint 1.

4. Acceptance Criteria (Given-When-Then)

Acceptance criteria define observable conditions for deciding whether a story meets its intended outcome. They are written in Given-When-Then form and should be checked during the sprint, not postponed until the final presentation. The selected Sprint 1 stories have explicit criteria below.

US-01: Register Field Zones

User story: As a farm manager, the user wants to register field zones and crop labels so that observations and decisions are associated with the correct area.

- Given an authorised user is creating a field zone, when the user supplies the required zone identifier and zone name, then the system records the zone and displays it in the zone list.
- Given a zone identifier already exists, when the user attempts to create another zone with the same identifier, then the system rejects the duplicate and explains the conflict.
- Given a zone has been saved, when a user opens the zone list, then the zone identifier, name, and crop label (if supplied) are shown consistently.

US-02: Associate Sensor Readings with Zones and Depths

User story: As a field operator, the user wants readings associated with a zone, depth, and timestamp so that soil conditions can be reviewed in context.

- Given a registered zone exists, when a reading with a known zone reference, depth label, timestamp, and moisture value is received, then the reading is associated with the correct zone and depth.
- Given a reading refers to an unknown zone or is missing a required field, when it is processed, then it is flagged for review and is not silently shown as a valid current reading.
- Given the same sample dataset is loaded twice in a controlled demonstration, when the same records are processed, then the zone and depth associations are repeatable and can be checked against the sample input.

Test	evidence	expected
For each criterion, retain the input example, the observed result, pass/fail status, and a brief note if a defect is found. An acceptance criterion is not satisfied solely because the screen looks complete.		

4. Acceptance Criteria (continued)

US-03: Identify Stale or Invalid Readings

User story: As a field operator, the user wants stale or invalid readings flagged so that unreliable information is not mistaken for current field conditions.

- Given a reading timestamp is older than the configured freshness window, when the validation step runs, then the reading is labelled stale and is excluded from generating a new irrigation recommendation.
- Given a moisture value is missing or outside the configured acceptable range, when the reading is checked, then it is labelled invalid and the user can see a reason.
- Given all readings for a zone are stale or invalid, when the recommendation view is opened, then the system shows that a reliable recommendation is unavailable instead of treating missing data as zero moisture.
- Given a valid newer reading arrives for a previously stale zone, when the record is validated, then the current valid reading is distinguishable from the older stale record.

US-04: View Multi-Depth Moisture Profile

User story: As a farm manager, the user wants to view moisture readings at different soil depths so that the decision is not based only on surface dryness.

- Given a selected zone has valid readings for at least two depth labels, when the zone profile is opened, then each depth and its moisture value are displayed with a reading timestamp.
- Given a depth has no valid reading, when the profile is displayed, then that depth is marked unavailable or missing; it is not displayed as a measured zero.
- Given a reading is marked stale or invalid, when the profile is displayed, then its status is visibly distinguished from valid readings.
- Given the user switches between two configured zones, when the profile changes, then the displayed readings correspond only to the selected zone.

Acceptance Review Rule

The Product Owner confirms whether the story outcome matches the agreed acceptance criteria. The development team supplies test evidence and explains known limitations. A story with failed critical criteria is not marked Done simply because most of its screen or workflow exists.

4. Acceptance Criteria for US-05

User story: As an irrigation operator, the user wants a rule-based recommendation for each zone with an explanation so that the next action is understandable and reviewable.

- Given a zone has valid, current moisture readings and configured decision thresholds, when the recommendation rule is evaluated, then the system returns a zone-specific status such as “irrigation indicated”, “hold”, or “review required” according to the configured rule.
- Given a recommendation is displayed, when the user inspects it, then the status, decision timestamp, and a concise reason based on the available inputs are visible.
- Given the readings required for a decision are stale or invalid, when a recommendation is requested, then the system does not issue an “irrigation indicated” outcome and instead states that review is required.
- Given the configured weather flag indicates a hold condition, when the rule is evaluated, then the recommendation reflects that hold condition rather than presenting a routine watering suggestion.
- Given a recommendation has been generated in Sprint 1, when the demonstration is reviewed, then no physical valve is activated; the increment provides a software recommendation only.

Important boundary

The Sprint 1 rule is a decision-support rule, not proof of an agronomically optimal irrigation policy. Crop-specific thresholds, weather rules, and microbial indicators require domain review and later validation before physical field use.

5. Product Backlog Prioritisation

The Product Owner orders the Product Backlog using business value, user need, dependency, risk reduction, and readiness. MoSCoW labels are used as a clear prioritisation aid: Must, Should, Could, and Won’t for the current sprint. The label “Won’t” means deferred from the current sprint, not excluded from the long-term product vision.

5.1 Prioritisation Rules

- Must: necessary to demonstrate the smallest safe, useful end-to-end path for Sprint 1.
- Should: important value but not required to validate the first data-to-recommendation flow.
- Could: valuable advanced capability that depends on stable data history or domain evidence.
- Priority may change after stakeholder review, but scope changes during a sprint should not silently invalidate the Sprint Goal.

5.2 Prioritised Product Backlog

The table provides the initial order for discussion. Priority and estimates are provisional and should be reviewed by the Product Owner and development team during backlog refinement.

Rank	ID / Story summary	Priority	Value / dependency rationale	SP (initial)
1	US-01 - Register field zones	Must	Foundation: all readings and recommendations need an identifiable zone.	3
2	US-02 - Associate readings to zone/depth/time	Must	Enables the system to represent relevant telemetry for each zone.	5
3	US-03 - Flag stale/invalid readings	Must	Reduces risk of decisions based on unreliable inputs; dependency for recommendations.	3
4	US-04 - View multi-depth moisture profile	Must	Directly addresses shallow-sensor limitation and creates visible user value.	3
5	US-05 - Explain rule-based recommendation	Must	Completes a demonstrable decision-support path after data quality is checked.	5
6	US-08 - Record recommendation audit history	Should	Improves traceability; most useful once recommendation events exist.	3
7	US-06 - Weather-based hold/warning	Should	Adds climate awareness; requires agreement on signals and hold rules.	5
8	US-11 - Operator hold/override	Should	Supports human oversight; authority and safety rules need stakeholder agreement.	3
9	US-07 - Local autonomy during network loss	Should	Important to long-term resilience; requires a distinct interruption/recovery test set.	8
10	US-12 - Role-based permissions	Should	Supports responsible operation; role boundaries need to be specified.	5
11	US-09 - Water/nutrient trend summaries	Could	Requires consistent history and agreed calculation definitions.	5
12	US-10 - Crop-stress indicator	Could	Depends on relevant historic data and agronomic validation.	8

5.3 Backlog Comparison: Immediate Value versus Deferred Scope

Sprint 1 commitment	Deferred backlog items	Reason for separation
Zone setup; simulated readings; stale/invalid checks; multi-depth profile; transparent recommendation	Weather integration, offline recovery, audit history, advanced prediction, role permissions	The first slice validates essential dependencies before adding integrations and higher-risk behaviour.

Sprint 1 commitment	Deferred backlog items	Reason for separation
Demonstration data and acceptance evidence	Claims of water savings, yield improvement, or microbiome preservation	Benefits require agreed baselines, reliable field data, and measured validation.

6. Sprint Backlog and Sprint Planning

6.1 Sprint 1 Planning Decision

Sprint 1 selects US-01 through US-05 for an initial total of 19 story points. The sequence follows the dependency chain from zone identity to validated readings, visible profiles, and an explainable recommendation. Work is selected as a coherent increment rather than as five unrelated features.

6.2 Editable Scrum Workflow Diagram

The workflow below is built from editable Word table cells, not an embedded image. The cycle emphasises inspection and adaptation, as required by the Unit 1 Scrum and Agile project-management topics.

1 Stakeholder discovery and product vision **Confirm the most valuable problem and intended outcomes.**
↓
2 Product Backlog refinement **Write epics and testable user stories; clarify dependencies.**
↓
3 Sprint Planning **Agree Sprint Goal, select stories, estimate effort, and break work into tasks.**
↓
4 Sprint execution and Daily Scrum **Collaborate, inspect progress toward the goal, and raise impediments.**
↓
5 Acceptance checks and Sprint Review **Demonstrate the increment and collect stakeholder feedback.**
↓

6 Sprint Retrospective **Identify one or two process improvements for the next sprint.**
↺ Review feedback updates Product Backlog priorities **The next cycle begins with refined needs and lessons learned.**

6.3 Sprint Planning Questions

- Why is this sprint valuable? It validates the smallest end-to-end path from a zone record to a reasoned irrigation recommendation.
- What can be Done by the end? The five selected stories meeting their acceptance criteria, supported by a demonstration and test evidence.
- How will the work be completed? The team decomposes each story into tasks, coordinates daily, checks criteria continuously, and raises blockers early.

6.4 Sprint Backlog: Actionable Tasks

The tasks below are implementation-neutral work items. They describe the outcome to be performed rather than naming a language, framework, database, broker, or hardware platform. Task ownership should be agreed by the cross-functional team during Sprint Planning.

Story	Task ID	Actionable task	Expected evidence / completion check
US-01	T1.1	Confirm required zone fields and identifier rules with the Product Owner.	Story notes and field list agreed.
US-01	T1.2	Prepare two sample zones with distinct identifiers and crop labels for the demonstration.	Sample zones can be opened and distinguished.
US-01	T1.3	Define and check duplicate-identifier behaviour.	Duplicate scenario fails with a clear message.
US-01	T1.4	Walk through the zone list with a representative user.	Feedback recorded and acceptance criteria checked.
US-02	T2.1	Agree the minimum reading record: zone, depth, timestamp, moisture value, and optional context fields.	Reading record definition documented.
US-02	T2.2	Create a repeatable sample set with readings for more than one depth.	Input data includes zone and depth labels.
US-02	T2.3	Define how a reading is associated with a registered zone.	Known and unknown zone cases can be tested.
US-02	T2.4	Check that incoming readings retain their timestamps and depth labels.	Displayed record can be traced to sample input.

Story	Task ID	Actionable task	Expected evidence / completion check
US-02	T2.5	Review record completeness with the test responsibility.	Missing-field scenario is included in test notes.
US-03	T3.1	Agree a configurable freshness window and valid-value boundaries for the demonstration.	Assumptions documented and approved for prototype testing.
US-03	T3.2	Prepare stale, missing, and out-of-range sample readings.	Each invalidity type has a reproducible input.
US-03	T3.3	Check that invalid/stale readings are labelled and excluded from new recommendations.	Acceptance criteria pass or defect is logged.
US-03	T3.4	Verify that a zone with no valid reading reports an unavailable/review state.	No missing measurement is silently treated as zero.

Task Management Practice

Each task should be visible on the team task board with a clear status, an agreed owner or pair, and a testable completion condition. Work in progress should be limited so that the team finishes started stories instead of beginning every story simultaneously. If scope is at risk, the team should raise the issue early and collaborate with the Product Owner on options that preserve the Sprint Goal.

6.5 Sprint Backlog: Remaining Tasks

Story	Task ID	Actionable task	Expected evidence / completion check
US-04	T4.1	Agree a readable way to identify zones, depths, values, timestamps, and data status.	User can interpret the profile in a short walkthrough.
US-04	T4.2	Prepare at least two depth readings for each of the sample zones.	Sample set supports multi-depth display.
US-04	T4.3	Check valid, stale, invalid, and missing-depth display states.	Each state has a visible distinction.
US-04	T4.4	Conduct a task-based review: ask a user to identify the deeper reading and its status.	Observation and corrections recorded.
US-05	T5.1	Agree the demonstration-only decision rules and the meaning of each status.	Rules documented; agronomic thresholds marked as configurable.
US-05	T5.2	Define how validated readings and the configured weather flag affect a recommendation.	Decision table includes valid and invalid data cases.
US-05	T5.3	Ensure each recommendation includes zone, status, timestamp, and reason.	Displayed recommendation satisfies criteria.
US-05	T5.4	Test at least one irrigate-indicated, one hold/review, and one invalid-data scenario.	Pass/fail evidence captured for each scenario.
US-05	T5.5	Confirm the prototype does not actuate physical valves in Sprint 1.	Review script demonstrates software advice only.

Story	Task ID	Actionable task	Expected evidence / completion check
All	T6.1	Run acceptance checks for all five selected stories and record results.	Traceable checklist with outcomes.
All	T6.2	Prepare a short demonstration path covering normal and exception scenarios.	Sprint review can be completed without improvising data.
All	T6.3	Review incomplete items, defects, and feedback with Product Owner.	Accepted, rejected, and deferred work is documented.

6.6 Scrum Events and Expected Inputs/Outputs

Event	Purpose in this sprint	Output
Sprint Planning	Agree value, Sprint Goal, selected stories, estimates, and initial task plan.	Sprint Goal and Sprint Backlog
Daily Scrum	Inspect progress toward Sprint Goal and adapt the near-term plan.	Updated task state and surfaced impediments
Sprint Review	Demonstrate the increment to stakeholders and gather practical feedback.	Acceptance decisions and new backlog insights
Sprint Retrospective	Discuss collaboration, workflow, and quality practices.	Small improvement action for next sprint
Backlog refinement	Clarify future stories and ordering as new information appears.	Better-prepared future Product Backlog items

7. Effort Estimation Using Story Points

7.1 Estimation Technique

The team uses relative estimation with a Fibonacci-style sequence (1, 2, 3, 5, 8, 13). Story points represent a comparative judgement of effort, complexity, uncertainty, and coordination. They are not hours or person-days. Planning Poker is suitable: team members select estimates independently, discuss large differences, and agree a shared estimate after assumptions and dependencies are understood.

Points	Interpretation for this project
1-2	Small, familiar, limited ambiguity; one clear outcome and few dependencies.
3	Moderate, well understood, with several testable conditions.
5	Several related activities or edge cases; moderate uncertainty or cross-role coordination.

Points	Interpretation for this project
8	Substantial integration, unresolved dependencies, or several distinct failure scenarios.
13	Too large or uncertain for reliable sprint commitment; split before selection where possible.

7.2 Sprint 1 Estimates and Justification

Story	Points	Rationale	Dependency / uncertainty
US-01 - Register zones	3	Limited configuration behaviour with clear duplicate checking and sample records.	Low; provides the identity foundation.
US-02 - Associate readings	5	Includes reading record definition, zone/depth association, timestamp retention, and repeatable samples.	Depends on US-01; moderate data-quality coordination.
US-03 - Validate readings	3	Several clear cases: stale, missing, out-of-range, and no valid record.	Depends on US-02; boundary conditions must be agreed.
US-04 - Multi-depth profile	3	Display and distinguish valid, stale, invalid, and missing readings by depth.	Depends on US-02 and US-03; usability review required.
US-05 - Recommendation	5	Combines validated conditions, configurable rules, weather hold, explanation, and negative test cases.	Depends on US-03 and US-04; agronomic rule validity remains a limitation.
Total planned	19	Initial planning total for the proposed two-week Sprint 1.	Reconfirm at planning based on real team capacity.

7.3 Why the Estimates Are Provisional

No historical velocity, confirmed team size, or actual skill distribution is provided in the assessment brief. Therefore, 19 story points are a proposed baseline, not a guaranteed commitment. The team should compare the selected work with its available capacity during Sprint Planning and reduce scope if the estimate exceeds its realistic ability to meet the Definition of Done. A story should not be compressed into fewer points merely to satisfy a target.

8. Sprint Goal and Expected Deliverables

8.1 Sprint Goal

> **Sprint Goal**
>
> Deliver a demonstrable first-sprint software increment that represents multiple field zones, accepts simulated soil readings at different depths, identifies stale or invalid data, displays validated moisture profiles, and produces transparent rule-based irrigation recommendations without operating physical valves.

8.2 Expected Deliverables at Sprint Review

- A zone register containing at least two sample field zones, each with a distinct identifier.
- A repeatable set of simulated readings linked to zone, depth, and timestamp.
- Visible validation results for current, stale, missing, and out-of-range readings.
- A multi-depth moisture profile that clearly distinguishes measured values from unavailable or invalid values.
- A zone-level recommendation showing status, timestamp, and a human-readable reason based on configured demonstration rules.
- A traceable acceptance checklist with passed tests, failed tests, and unresolved defects recorded.
- A short demonstration script and stakeholder feedback log to inform backlog refinement.

8.3 Definition of Done for Sprint 1

- The story acceptance criteria have been checked and evidence recorded.
- Normal and exception scenarios have been tested using repeatable sample data.
- The user-facing status and messages are understandable in a stakeholder walkthrough.
- Known limitations, untested areas, and deferred items are documented.
- The increment is reviewable by the Product Owner and stakeholders without implying that it is production-ready or field-validated.

8.4 Risks and Responses

Risk	Effect on sprint	Planned response
Unclear moisture thresholds	Recommendation may appear more authoritative than evidence allows	Use configurable demonstration thresholds; label them as provisional and request agronomic review.
Invalid data not considered early	Unsafe or misleading recommendation state	Prioritise US-03 before US-05 and include stale/missing data tests.

Risk	Effect on sprint	Planned response
Too much scope in first sprint	Incomplete features and weak acceptance evidence	Maintain 19-point provisional scope; defer advanced prediction, physical valves, and production connectivity.
Stakeholders unavailable for review	Usability gaps may remain unnoticed	Arrange review early; use documented questions and mark feedback still needed.
Claims exceed evidence	Misleading environmental or yield claims	Report only measured prototype behaviour; field benefits remain hypotheses until evaluated.

8.5 Review, Measurement, and Adaptation

The Sprint Review should assess the increment against the Sprint Goal rather than count the number of screens or tasks completed. Stakeholders should be able to follow at least one normal case and multiple exception cases. Review findings should be converted into backlog changes, not treated as informal comments that disappear after the meeting.

Review question	Evidence to inspect	Possible backlog response
Can a user identify the correct zone and depth?	Zone setup and profile walkthrough	Improve zone labels or profile layout if the wrong value is easy to select.
Can unreliable data be recognised?	Stale, invalid, missing, and valid-data scenarios	Clarify thresholds, freshness policy, or status language.
Can a user explain the recommendation?	Recommendation status and reason	Refine rule wording or add missing input context.
Are the rules agronomically appropriate?	Review by farm or soil specialist	Keep as demonstration rule or add domain validation work.
Can the team complete work with less rework?	Task flow, defects, and retrospective notes	Improve story readiness, test design, or collaboration habits.
Are expected environmental outcomes measurable?	Agreed baseline and list of data to collect	Create a future evaluation story before claiming savings.

8.6 Indicative Follow-up Sprint Themes

Later sprint planning should be based on stakeholder feedback and the team's demonstrated capacity. Candidate themes include:

1. Improve auditability by recording recommendation history and the input conditions behind each decision.
2. Add an agreed weather-related hold rule and test it against contradictory or missing weather data.
3. Simulate network interruption, local rule availability, recovery, and later synchronisation as a separate resilience increment.

4. Define authorised override behaviour and clear operator responsibilities for uncertain conditions.
5. Establish a baseline and a measurement plan for water utilisation and nutrient-retention indicators before reporting benefits.

Agile management principle

The Product Backlog remains adaptable, but adaptations should be evidence-based. The Sprint Review informs what to do next; the Retrospective informs how the team should work better next time.

Conclusion

This workshop translates the allocated irrigation network into a prioritised, testable first-sprint plan. The long-term vision remains water conservation, multi-zone monitoring, soil-health awareness, weather context, and resilient operation. Sprint 1 is limited to zone registration, simulated telemetry, data validation, multi-depth visibility, and transparent recommendations.

The Scrum approach supports this work through a clear product vision, Product Owner-led backlog ordering, collaborative story estimation, actionable sprint tasks, daily coordination, acceptance checks, stakeholder review, and retrospective improvement. Explicit assumptions and deferred scope reduce the risk of making unrealistic commitments. The plan also recognises that agronomic thresholds, physical valve control, microbial indicators, water savings, and crop-yield benefits require further validation and cannot be established by the first software demonstration alone.

The proposed Sprint Goal, 19-point initial estimate, acceptance criteria, task breakdown, and Definition of Done provide a concrete basis for a student project team to begin work and review progress. The plan should be adjusted after the real team confirms its capacity and stakeholders clarify field-specific requirements.

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